

ECE 565 Weekly Progress Report

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Project: SCRE-Credit: Stability- and Cost-Regularized Ensemble for Credit Risk

Scope change from proposal: German Credit / Option A is archived. Active scope is Taiwan default prediction with HELOC external validation. HELOC Optuna is still pending as a long-running job.

Evidence From Current Artifacts

Item	Result	Comment
Taiwan data audit	30,000 rows, 40 model columns, default rate 22.12%	Missing values and duplicate full rows reported; negative bill amounts are retained and audited.
HELOC data audit	9,871 rows, 31 model columns, bad-flag rate 52.03%	Special negative codes are documented through the cleaning-decisions table.
Taiwan RF baseline test	Acc 0.818, F1 0.468, ROC-AUC 0.780, PR-AUC 0.560	Untuned baseline; test used only for final evaluation rows.
Taiwan CatBoost test	Acc 0.818, F1 0.466, ROC-AUC 0.785, PR-AUC 0.565	Best current untuned boosting result by PR-AUC on Taiwan.
SCRE-Credit Taiwan	t=0.50: Acc 0.815, F1 0.523, ROC-AUC 0.783, PR-AUC 0.565	Cost threshold 0.25 cuts FN=5/FP=1 expected cost from 3980 to 3338 (16.1% improvement).
SCRE-Credit HELOC	t=0.50: Acc 0.732, F1 0.747, ROC-AUC 0.807, PR-AUC 0.813	External validation is promising, but the best cost threshold 0.10 uses very high recall and low specificity; this needs careful interpretation.
Reproducibility check	pytest: 26 passed; environment versions exported.	LightGBM feature-name warning was fixed by preserving pandas output through sklearn pipelines.

Literature And Current Results

Model / Paper	Method	Accuracy	F1	AUC	Other	Best?	Notes
Alam et al. (2020)	GBDT + K-means SMOTE	~0.887	NR	NR	Reported high	No	Proposal benchmark; likely not directly

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	on Taiwan				accuracy		comparable until split/CV protocol is matched.
K-means SMOTE + BP NN (2021)	Class imbalance handling + neural net	NR	NR	0.929	AUC improved 0.765 to 0.929	No	Very strong AUC claim; must be checked for protocol differences before comparing.
Interpretable DL + XAI (2023)	Deep model with XAI	0.835	NR	NR	Sensitivity 0.882, specificity 0.988	No	Included from proposal; current work is not yet claiming to exceed it.
DNN empirical study (2025)	Deep neural network on Taiwan	0.818	0.482	0.770	G-mean 0.603	No	Closest to current Taiwan metrics; our SCORE cost/recall setting has higher F1 but lower accuracy at cost-optimal threshold.
Lin & Wang (2025)	100 XGBoost seeds + SHAP stability	0.795	NR	0.777	Kendall W ~0.98 all features	No	Most relevant new benchmark for planned SHAP stability work.
MY Baseline Model	Random Forest, train-fitted preprocessing	0.818	0.468	0.780	PR-AUC 0.560	No	Honest baseline; no hyperparameter tuning.
MY Proposed Model	SCORE-Credit ensemble + cost thresholding	0.815	0.523	0.783	PR-AUC 0.565; cost -16.1%	Partial	Promising on cost- sensitive decision policy, but not yet a final claim because full stability and ablation tests remain.

Next Week Plan

1. Run HELOC Optuna 50-trial LightGBM job with 5x3 repeated stratified CV. Deliverable: Updated best_params.csv / optuna_trials.csv with HELOC results and saved study joblib.
2. Run the full SHAP stability experiment with 50 seeds for the selected final model family. Deliverable: shap_stability_rankings.csv, stability summary plots, and a no-overclaim paragraph for the report.
3. Add SCORE-Credit ablation: without cost term, without calibration term, without stability/faithfulness proxy. Deliverable: Ablation table showing which term changes PR-AUC, recall, Brier, and expected cost.